# Caelan Dunlea

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## EDUCATION

**Bachelor of Science in Electrical Engineering,** May 2029
The University of Alabama,Tuscaloosa, AL
Minor in Mathematics, Randall Research Scholars Program
GPA: 4.00/4.00

- National Merit Finalist & Scholarship Recipient

## EXPERIENCE

**Staff Engineer Intern,** Reliable Microsystems, Franklin, Tennessee, May 2026 - Aug 2026

- Designed and routed a 7"x8" PCB for semiconductor technology characterization and harsh environment testing, with a focus on electrical isolation, bias control, and automated measurement
- Completed layouts in Cadence Virtuoso and performed top-level integration across 3 chip sides on 2 chips, including custom routing and development of routing/dummy-routing cells
- Wrote python scripts and Verilog to support development and validation of an RTL to GDS design flow

**Electrical & Outreach Team Member,** Alabama Astrobotics, Tuscaloosa, Alabama, Aug 2025 - Present

- Strategized implementation and placement of electrical network for NASA Lunabotics Competition robot, combining LiDAR, IMU, stereo cameras, CAN bus, and VESC controllers
- Build electrical system for a 3×3 ft triangular interactive LED board featuring 1,280 LEDs, 6 full-size buttons, and 3 microcontrollers; safe power distribution, electrical dampening, and modular design
- Formulate schematics for 5 custom PCBs with KiCad software, implementing PCBs with broader LED system for full board control

**Electrical Engineer, BDU Team,** Battery Workforce Challenge, Tuscaloosa, Alabama, Aug 2025 - Apr 2026

- Modeled 2950 board and system components in Fusion 360 and Altium Designer for high-accuracy current tracking on-board shunt resistor, battery stack voltage, link voltage, and pre-charge states monitoring
- Collaborated with a 13-member multidisciplinary team to integrate high-voltage harness into full system architecture, managing design documentation for Department of Energy competition where team placed 3rd overall

**Undergraduate Researcher in Embedded Systems,** Dr. Sazonov's Lab, Tuscaloosa, Alabama, Jan 2026 - Present

- Develop Python scripts and machine learning methods to analyze accelerometer and optical sensor data against annotated eating events to predict food intake detection for the Automatic Ingestion Monitor device
- Research and test machine learning models (CNNs and Transformers), optimizing parameters for 90% accuracy

**Lab Manager,** Randall Research Scholars Program, Tuscaloosa, Alabama, Jan 2026 - Present

- Deploy and manage applications by operating SCCM, Active Directory, and Synology NAS for 180-person lab
- Maintain lab hardware and software, including security updates, new version implementation, account creation, and upkeep of a 20TB lab infrastructure
- Administer Linux and Windows backend systems, troubleshooting configuration, storage, and backup issues

## SKILLS & QUALIFICATIONS

Electrical & Design: KiCad, Altium Designer, Fusion 360, Solidworks, Cadence Virtuoso, Cadence OrCAD
Software Tools: Linux, Git, Docker, SCCM, Active Directory, VSCode
Programming: Python, C/C++, MATLAB, SQL, HTML, Javascript, C#, CSS, Verilog, Fortran, RTL to GDS flow
Data & Simulation: PyTorch, NumPy, Jupyter Notebooks, Anaconda, MATLAB, Microsoft Office Suite

## AWARDS AND RECOGNITIONS

Darren Evans-Young Outstanding Freshman Award (2026) · Deans Scholar (2025-2026) · National Merit Scholarship Recipient (2025) · CREATE Award at VEX Worlds (2025) · VEX Robotics State Champions (2025) · All-State Clarinetist (2025)